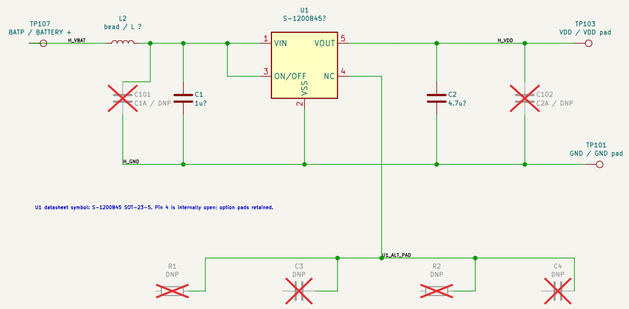
01 POWER / INPUT FILTER

H01: likely S-1200B45 LDO. Nominal 4.5 V is a candidate, not a measured rail.

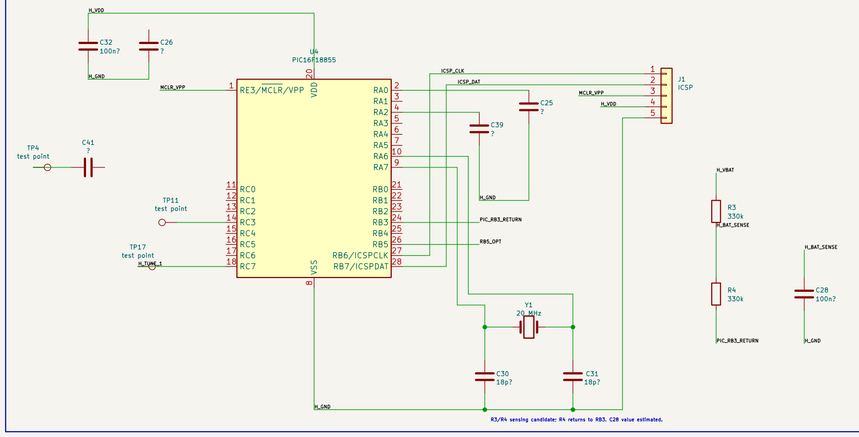


U1 datasheet symbol: S-1200B45 SOT-23-5. Pin 4 is internally open; option pads retained.

Optional R1/R2/C3/C4 network: empty footprints; free ends unresolved.

02 PIC CONTROLLER / CLOCK / ICSP

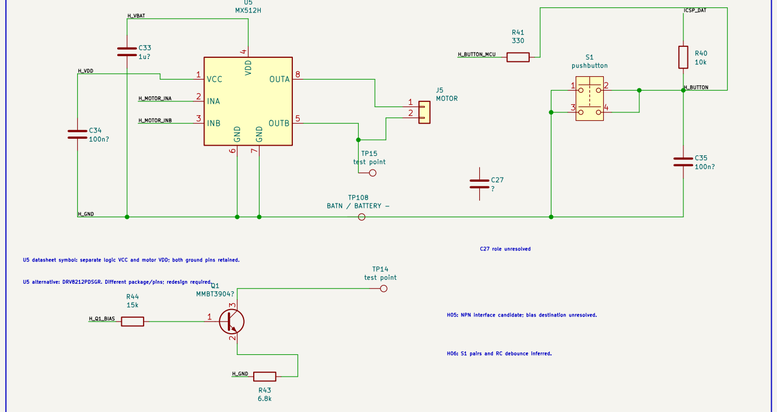
PIC local photo traces added: 14 GPIO pads open. Remote destinations partly inferred.



R3/R4 sensing candidate: R4 returns to RB3. C28 value estimated.

03 MOTOR / BUTTON / BATTERY INTERFACE

H04: MX512H typical supply/output topology. Motor A/B and MCU control assignments remain hypotheses.



US datasheet symbol: separate logic VCC and motor VDD; both ground pins retained.

US alternative: DRV8212PDSGR. Different package/pins; redesign required.

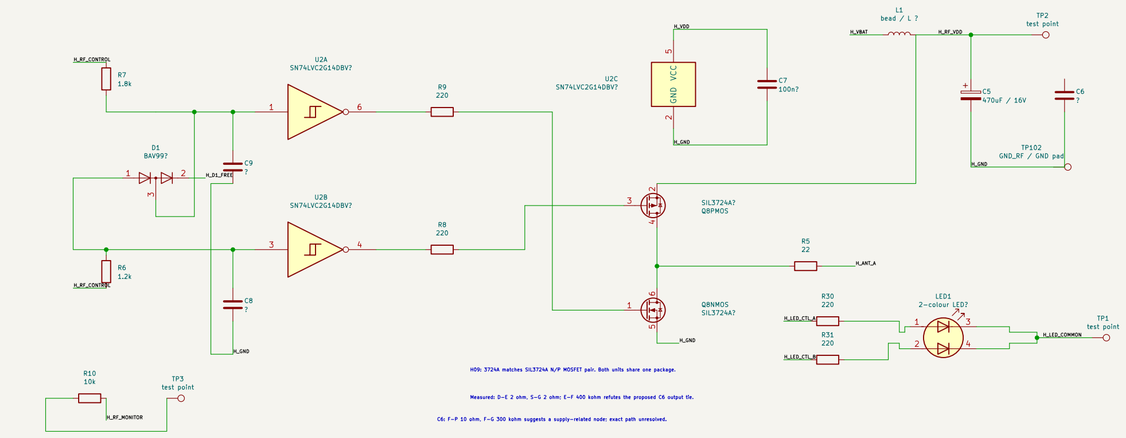
C27 role unresolved

H05: NPN interface candidate; bias destination unresolved.

H06: S1 pairs and RC debounce inferred.

04 ANTENNA EXCITATION / LOGIC / INDICATOR

H07: C14R inverter candidate, Q8 source rail inferred; C6 destination unresolved.



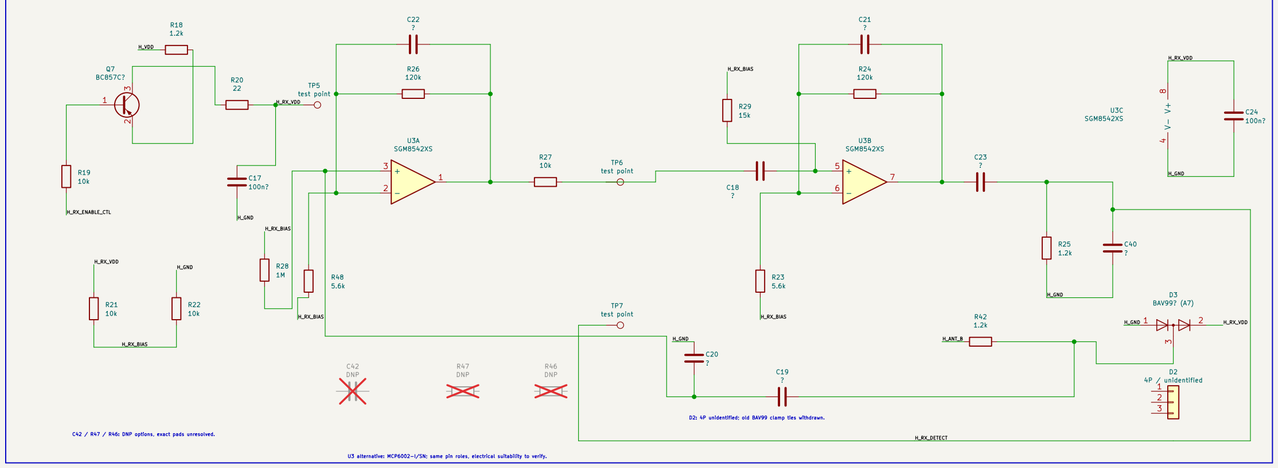
H09: 3724A matches SIL3724A N/P MOSFET pair. Both units share one package.

Measured: D–E 2 ohm, S–G 2 ohm; E–F 400 kohm refutes the proposed C6 output tie.

C6: F–P 10 ohm, F–G 300 kohm suggests a supply-related node; exact path unresolved.

05 RF RECEIVE / GAIN / ENVELOPE

H10–H12: working topology around photographed 120k feedback resistors. Filters, bias, and interstage routes inferred.



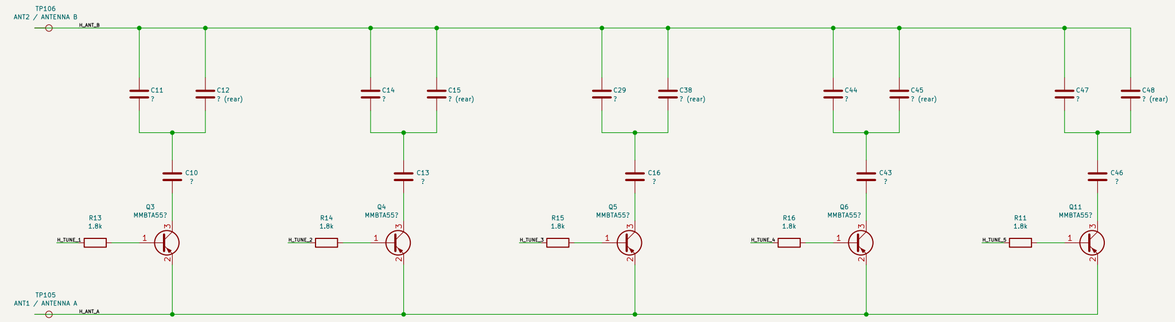
C42 / R47 / R46: DNP options, exact pads unresolved.

U3 alternative: MCP6002-I/SN: same pin roles, electrical suitability to verify.

D2: 4P unidentified: old BAV99 clamp ties withdrawn.

06 SWITCHED ANTENNA CAPACITANCE

H13: 2N devices are probably PNP BJTs (MMBTA55 family; PBSS51407 is an alternative). Capacitor values unknown.



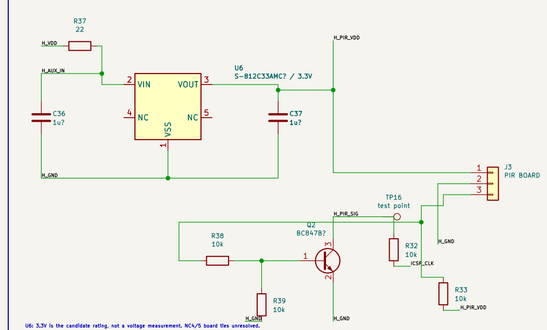
Selected working hypothesis: Ctop in series with (Cfront || Crear). Rear-bank vias suggest paired lower capacitors.

Ceff = Ctop * (Cfront + Crear) / (Ctop + Cfront + Crear). Equal values give 2C/3 per enabled branch.

All-parallel and all-series banks remain alternatives. Emitter rail / collector orientation is provisional.

07 PIR / AUXILIARY RAIL

H14–H15: C2N supports S-812C33AMC 3.3V LDO. C/R3 ground supported by resistance.



U6: 3.3V is the candidate rating, not a voltage measurement. NC4/5 board ties unresolved.

08 UNRESOLVED / UNPOPULATED OPTIONS

D4/D5 cathodes traced to ICSP; opposite vias open. D6 type and routes uncertain.



READING THIS WORKING RECONSTRUCTION

One sheet. Functional wiring is proposed; this is not a verified production circuit.

150 original main-board catalog entries, including DNP footprints and named pads.

H_* = inferred inter-block net. Other net names do not imply measurement.

? on a part value or pin function means an estimate or candidate identity.

Library pin numbers are mapped to photo pads in evidence/pin_crosswalk.json.

U2 and U3 use multiple symbol units on this same sheet, including supply units.

29 original local visual fragments are retained; added connections are hypotheses.

PIC GPIOs are intentionally unassigned where photographs cannot support a map.

DNP = unpopulated. An open pin means unresolved, not a certified no-connect.

PIR daughterboard internals are not yet traced; J3 represents its interface.

No capacitor value or rail voltage in this drawing was electrically measured.

See docs/RECONSTRUCTION.md for evidence, alternatives, and calculations.



Spare footprints / unassigned test pads

v0.7 | KiCad 10.0.5 | 2026-10-08

PIC photo traces corrected: C5 470uF / 16V. Remaining hypotheses documented.

KiCad 10 libraries + four datasheet symbols | MCP authoring

Sheet: /
File: PetSafe_1001339.kicad_sch

Title: PetSafe 100–1339 R03 A | single-sheet reconstruction

Size: A2	Date: 2026-10-08	**Rev: 0.7**
KiCad E.D.A. 10.0.5		Id: 1/1